

MLA0202-Fundamentals Of Machine Learning

Analytical Problem Solving

Question 1: Multiple Linear Regression

Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression
data = {
    'Area': [1000, 1500, 800, 1200, 2000],
    'Bedrooms': [2, 3, 2, 3, 4],
    'Price': [50, 75, 40, 60, 90]
}
df = pd.DataFrame(data)
X = df[['Area', 'Bedrooms']]
y = df['Price']
model = LinearRegression()
model.fit(X, y)
print("Intercept ( $\beta_0$ ):", model.intercept_)
```

```
print("Coefficient for Area ( $\beta_1$ ):", model.coef_[0])
print("Coefficient for Bedrooms ( $\beta_2$ ):", model.coef_[1])
new_house = [[1000, 2]]
prediction = model.predict(new_house)
print("Predicted Price:", prediction[0], "Lakhs")
```

Output:

```
Welcome linear regression.py X Untitled-1
```

```
Users > nandiniboyapati > Documents > ppt > linear regression.py > data
```

```
1 import pandas as pd
2 from sklearn.linear_model import LinearRegression
3 data = {
4     'Area': [1000, 1500, 800, 1200, 2000],
5     'Bedrooms': [2, 3, 2, 3, 4],
6     'Price': [50, 75, 40, 60, 90]
7 }
8 df = pd.DataFrame(data)
9 X = df[['Area', 'Bedrooms']]
10 y = df['Price']
11 model = LinearRegression()
12 model.fit(X, y)
13 print("Intercept ( $\beta_0$ ):", model.intercept_)
14 print("Coefficient for Area ( $\beta_1$ ):", model.coef_[0])
15 print("Coefficient for Bedrooms ( $\beta_2$ ):", model.coef_[1])
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [X] ... | [ ] [X]
```

```
/usr/local/bin/python3 "/Users/nandiniboyapati/Documents/ppt/linear regression.py"
```

```
nandiniboyapati@Nandinis-MacBook-Air / % /usr/local/bin/python3 "/Users/nandiniboyapati/Documents/ppt/linear r
```

```
egression.py"
```

```
Intercept ( $\beta_0$ ): 8.504672897196265
```

```
Coefficient for Area ( $\beta_1$ ): 0.042523364485981305
```

```
Coefficient for Bedrooms ( $\beta_2$ ): -0.28037383177570213
```

```
/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/sklearn/utils/validation.py:28
```

```
27: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
```

```
warnings.warn(
```

```
Predicted Price: 50.467289719626166 lakhs
```

Question 2: Linear Classification

Code:

```
import pandas as pd

data = {
    'Offer': [3, 2, 1, 3, 0],
    'Win': [2, 1, 0, 3, 1],
    'Spam': [1, 1, 0, 1, 0]
}

df = pd.DataFrame(data)
print(df)

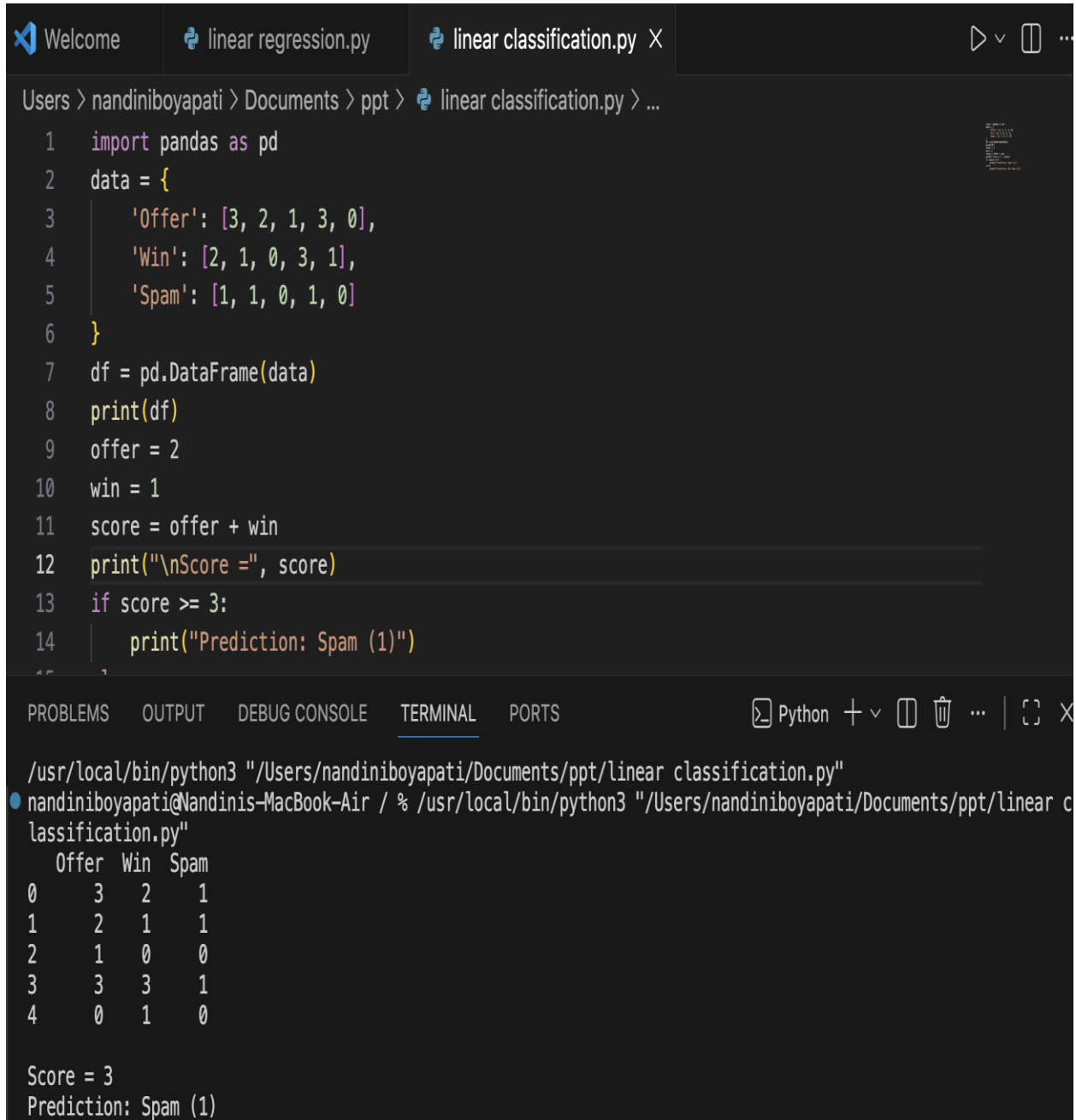
offer = 2
win = 1

score = offer + win

print("\nScore =", score)

if score >= 3:
    print("Prediction: Spam (1)")
else:
    print("Prediction: Not Spam (0)")
```

Output:



The image shows a VS Code editor window with a Python file named `linear classification.py`. The script imports `pandas` and creates a DataFrame with columns 'Offer', 'Win', and 'Spam'. It then calculates a score for each row based on the 'Offer' and 'Win' values. If the score is greater than or equal to 3, it predicts 'Spam (1)'. The terminal output shows the DataFrame contents and the final score and prediction for the first row.

```
1 import pandas as pd
2 data = {
3     'Offer': [3, 2, 1, 3, 0],
4     'Win': [2, 1, 0, 3, 1],
5     'Spam': [1, 1, 0, 1, 0]
6 }
7 df = pd.DataFrame(data)
8 print(df)
9 offer = 2
10 win = 1
11 score = offer + win
12 print("\nScore =", score)
13 if score >= 3:
14     print("Prediction: Spam (1)")
```

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```
/usr/local/bin/python3 "/Users/nandiniboyapati/Documents/ppt/linear classification.py"
nandiniboyapati@Nandinis-MacBook-Air / % /usr/local/bin/python3 "/Users/nandiniboyapati/Documents/ppt/linear c
lassification.py"
  Offer  Win  Spam
0      3    2     1
1      2    1     1
2      1    0     0
3      3    3     1
4      0    1     0

Score = 3
Prediction: Spam (1)
```

Question 3 : Naive Bayes Classification

Code:

```
]import pandas as pd
from sklearn.naive_bayes import BernoulliNB
data = {
    'Fever': [1, 1, 0, 1, 0],
    'Headache': [1, 0, 1, 1, 0],
    'Disease': [1, 1, 0, 1, 0]
}
df = pd.DataFrame(data)
X = df[['Fever', 'Headache']]
y = df['Disease']
model = BernoulliNB()
model.fit(X, y)
test = [[1, 0]]
prediction = model.predict(test)
probability = model.predict_proba(test)
print("Prediction:", "Yes" if prediction[0] == 1 else "No")
```

```
print("Probability:", probability)
```

Output :

```
Users > nandiniboyapati > Documents > ppt > naive bayes theorem.py > test
1 import pandas as pd
2 from sklearn.naive_bayes import BernoulliNB
3 data = {
4     'Fever': [1, 1, 0, 1, 0],
5     'Headache': [1, 0, 1, 1, 0],
6     'Disease': [1, 1, 0, 1, 0]
7 }
8 df = pd.DataFrame(data)
9 X = df[['Fever', 'Headache']]
10 y = df['Disease']
11 model = BernoulliNB()
12 model.fit(X, y)
13 test = [[1, 0]]
14 prediction = model.predict(test)

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/usr/local/bin/python3 "/Users/nandiniboyapati/Documents/ppt/naive bayes theorem.py"
nandiniboyapati@Nandinis-MacBook-Air / % /usr/local/bin/python3 "/Users/nandiniboyapati/Documents/ppt/naive bayes theorem.py"
/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/sklearn/utils/validation.py:28
27: UserWarning: X does not have valid feature names, but BernoulliNB was fitted with feature names
warnings.warn(
/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/sklearn/utils/validation.py:28
27: UserWarning: X does not have valid feature names, but BernoulliNB was fitted with feature names
warnings.warn(
Prediction: Yes
Probability: [[0.20661157 0.79338843]]
```